

Experiment No.	8
Experiment Title	Clustering Human Activity Recognition Data using K-Means, DBSCAN, and Hierarchical Clustering
Student Name	S.R.Aadhithya
Register Number	3122247001001
Faculty	Dr. Poreddy Ajay Kumar Reddy
Submission Date	20-09-2026

1 Objective

- To implement, evaluate, and analyze three distinct unsupervised clustering paradigms on high-dimensional sensor data from the Human Activity Recognition (HAR) dataset:
 - **Model A: K-Means Clustering** (centroid-based partitioning with Elbow and Silhouette optimization).
 - **Model B: DBSCAN** (Density-Based Spatial Clustering of Applications with Noise, exploring k -distance and grid search).
 - **Model C: Hierarchical Agglomerative Clustering (HAC)** (bottom-up agglomerative grouping comparing Ward, Complete, and Single linkage).
- To project high-dimensional smartphone sensor features onto a two-dimensional subspace via Principal Component Analysis (PCA) for visual verification.
- To evaluate clustering quality using both internal metrics (Silhouette Score, Davies–Bouldin Index, Calinski–Harabasz Index) and external ground-truth metrics (Adjusted Rand Index, Normalized Mutual Information).
- To analyze algorithm sensitivity, cluster geometry, and the curse of dimensionality across sensor-derived kinematics.

2 Problem Statement

Human Activity Recognition (HAR) utilizes triaxial acceleration and angular velocity signals captured by smartphone inertial sensors to classify physical behaviors into dynamic activities (*WALKING*, *WALKING_UPSTAIRS*, *WALKING_DOWNSTAIRS*) and static postures (*SITTING*, *STANDING*, *LAYING*). In this experiment, we explore unsupervised clustering without utilizing ground-truth labels during model fitting. We investigate how partitioning algorithms (K-Means), density-based methods (DBSCAN), and agglomerative hierarchical models group 561-dimensional kinematic observations, determine optimal cluster counts using the Elbow and Silhouette criteria, and examine the impact of high-dimensional distance concentration on density-based noise detection and hierarchical linkage criteria.

3 Dataset Description

Dataset Name	Human Activity Recognition Using Smartphones Dataset
Dataset Source	UCI Machine Learning Repository (Dataset ID: 240)
Number of Samples	10,299 total observations (Train: 7,352, Test: 2,947)
Number of Features	561 time and frequency domain continuous features
Sensor Modalities	3-axis Accelerometer and 3-axis Gyroscope sampled at 50 Hz
Number of Target Activities	6 activities (Walking, Upstairs, Downstairs, Sitting, Standing, Laying)
Missing Values	0 (Clean, fully preprocessed sensor signal windows)
Dimensionality Reduction	PCA 95% explained variance (retains 104 components) for DBSCAN/HAC

4 Exploratory Data Analysis

Exploratory analysis investigates the distribution of activities across the 10,299 recorded instances and projects the full 561-dimensional sensor space onto two principal components.

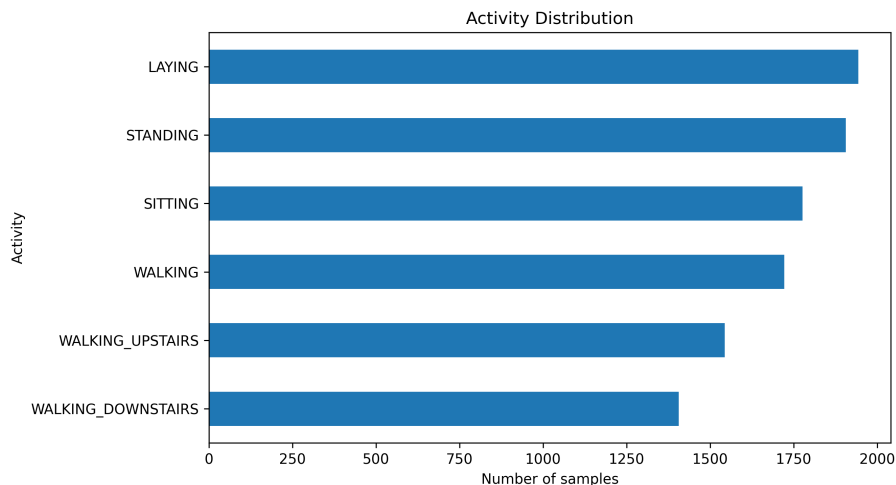


Figure 1: Ground Truth Activity Distribution in HAR Dataset

Inference: The dataset contains 10,299 instances across six activity categories: Laying (1,944), Standing (1,906), Sitting (1,777), Walking (1,722), Walking Upstairs (1,544), and Walking Downstairs (1,406). The class distribution is well-balanced across all physical regimes, ensuring that cluster evaluations are not biased by overwhelming majority classes.

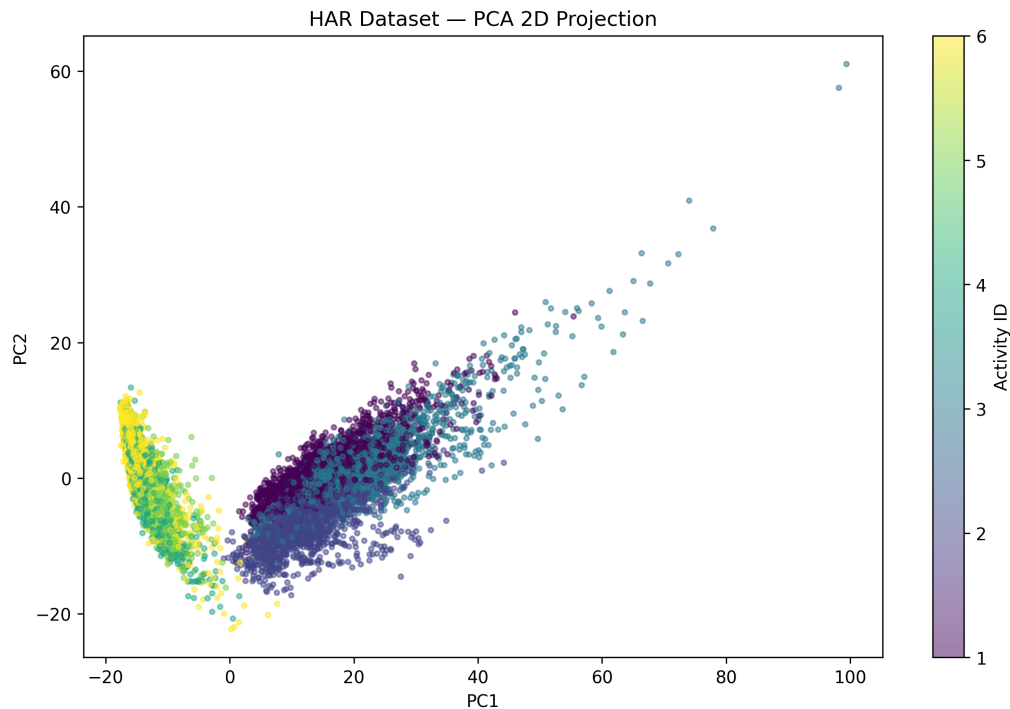


Figure 2: HAR Dataset Ground Truth — PCA 2D Projection

Inference: The 2D PCA projection reveals a clear macro-level structural bifurcation: dynamic activities (Walking, Walking Upstairs, Walking Downstairs) form a dense, contiguous cluster on the left side of the projection (negative PC1), whereas static postures (Sitting, Standing, Laying) form a distinct manifold on the right side (positive PC1). However, within each macro-group (e.g., Sitting vs. Standing), severe class overlap occurs, highlighting the challenge for unsupervised partitioning algorithms.

5 Data Preprocessing

- **Missing Values and Consistency:** Checked for missing or duplicate entries; the dataset is complete with zero null entries across all 10,299 records and 561 columns.
- **Feature Standardization:** Applied `StandardScaler` to normalize all 561 continuous features to zero mean and unit variance ($\mu = 0, \sigma = 1$), ensuring uniform Euclidean distance weighting across accelerometer and gyroscope metrics.
- **Dimensionality Reduction:** Performed Principal Component Analysis (PCA) retaining 95% of total dataset variance, compressing 561 features into 104 orthogonal principal components. This step removes noise, preserves kinetic information, and makes distance computations computationally tractable for $O(N^2)$ hierarchical clustering and DBSCAN.
- **Stratified Sampling for Distance Matrix Computation:** A stratified sample of 2,500 instances was sampled preserving activity label ratios to execute nearest-neighbor queries and hierarchical matrix decomposition reproducibly.

6 Machine Learning Algorithms for Clustering

6.1 K-Means Clustering

K-Means partitions n samples into k clusters by minimizing the within-cluster sum of squares (WCSS / Inertia):

$$J = \sum_{j=1}^k \sum_{x_i \in C_j} \|x_i - \mu_j\|^2 \quad (1)$$

Centroids are initialized using the **k-means++** heuristic to ensure rapid convergence, and clusters are iteratively updated until centroid shifts fall below the tolerance threshold.

6.2 DBSCAN (Density-Based Spatial Clustering of Applications with Noise)

DBSCAN defines clusters as continuous regions of high density separated by regions of low density. For each point p , if the ε -neighborhood $N_\varepsilon(p) = \{q \in D \mid \text{dist}(p, q) \leq \varepsilon\}$ satisfies $|N_\varepsilon(p)| \geq \text{min_samples}$, p is marked as a core point and an interconnected density component is grown. Unreachable points are labeled as noise (cluster -1).

6.3 Hierarchical Agglomerative Clustering (HAC)

Hierarchical clustering builds a nested hierarchy of clusters in a bottom-up fashion. Starting with each sample in its own singleton cluster, pairs of clusters are merged based on linkage criteria:

- **Ward's Linkage:** Minimizes total within-cluster variance increase ΔESS upon merging:

$$d(A, B) = \frac{n_A n_B}{n_A + n_B} \|\mu_A - \mu_B\|^2 \quad (2)$$

- **Complete Linkage:** Merges clusters minimizing the maximum pairwise distance: $d(A, B) = \max_{x \in A, y \in B} \|x - y\|$.
- **Single Linkage:** Merges clusters minimizing the minimum pairwise distance: $d(A, B) = \min_{x \in A, y \in B} \|x - y\|$.

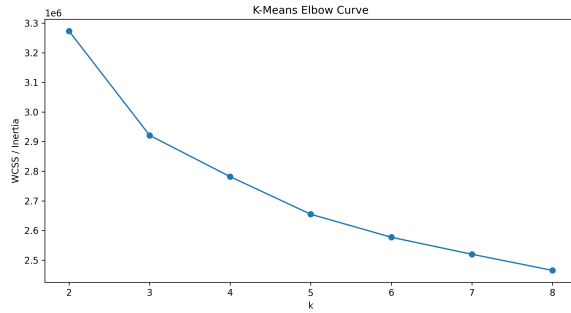
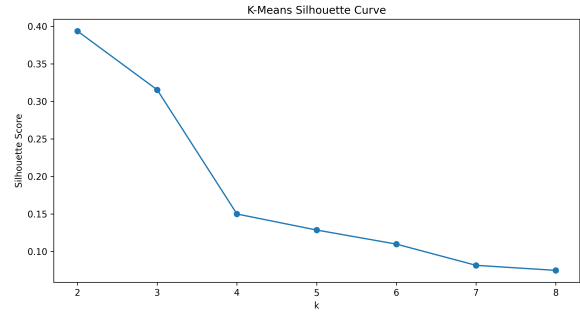
7 Experimental Results and Performance Analysis

7.1 K-Means Elbow Method and Silhouette Analysis

We evaluated K-Means clustering across cluster counts $k \in \{2, 3, 4, 5, 6, 7, 8\}$ on the full standardized 10,299-sample feature space. Table 1 reports the exact WCSS and Silhouette scores obtained from execution.

Table 1: K-Means Elbow Method and Silhouette Score Results

Number of Clusters (k)	WCSS (Inertia)	Silhouette Score
2	3272856.62	0.3937
3	2921077.60	0.3155
4	2781602.14	0.1500
5	2654789.79	0.1286
6	2577180.43	0.1099
7	2519774.22	0.0816
8	2465378.96	0.0749

Figure 3: K-Means Elbow Curve (k vs. WCSS)Figure 4: K-Means Silhouette Curve (k vs. Silhouette)

Inference: The elbow curve exhibits its primary reduction bend between $k = 2$ and $k = 3$, with diminishing returns in WCSS reduction for $k \geq 4$. Correspondingly, the Silhouette Score peaks decisively at $k = 2$ (0.3937), followed by $k = 3$ (0.3155), and drops sharply to below 0.15 for $k \geq 4$. This demonstrates that the feature space is intrinsically dominated by a 2-cluster geometric partition separating active dynamic movements from stationary postures. Therefore, $k = 2$ is chosen as the optimal partitioning parameter.

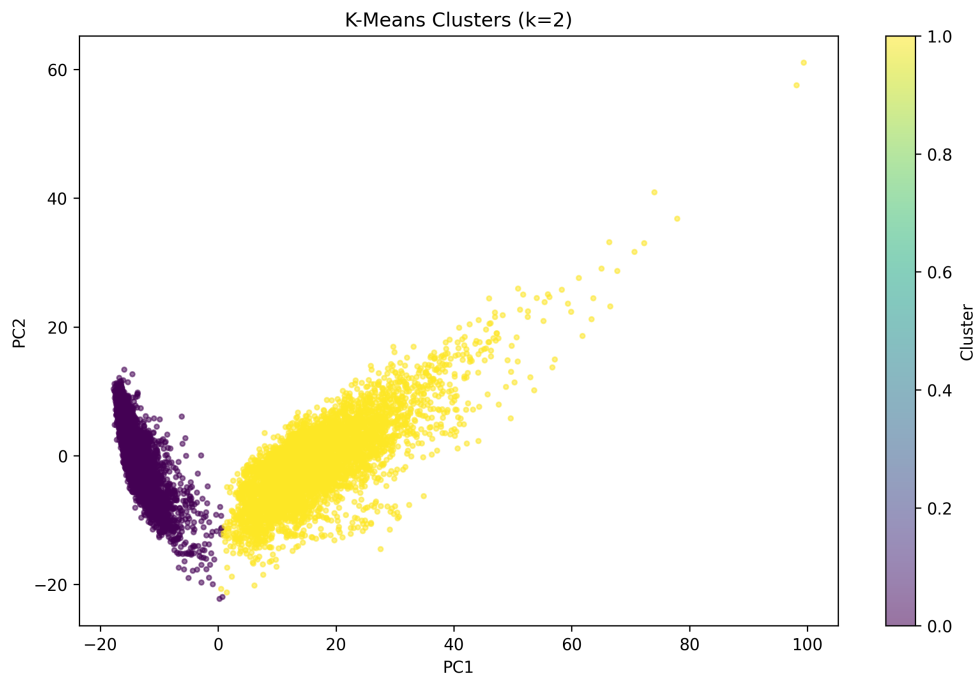


Figure 5: K-Means Clusters ($k = 2$) Projected in 2D PCA Space

Inference: The $k = 2$ K-Means clustering cleanly segregates the data into two well-separated spatial components: Cluster 0 encapsulates the dynamic motion observations (Walking, Stairs), while Cluster 1 captures the static postures (Sitting, Standing, Laying), matching the intrinsic structural boundary observed in EDA.

7.2 DBSCAN Density Analysis and Parameter Tuning

To determine an appropriate neighborhood radius ϵ , a 10-nearest neighbor distance graph was computed over the 95% PCA feature space.

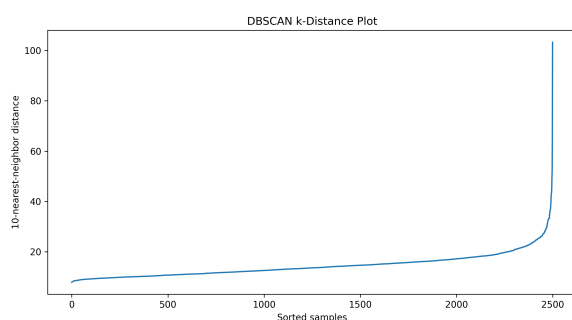


Figure 6: DBSCAN 10-Nearest Neighbor Distance Plot

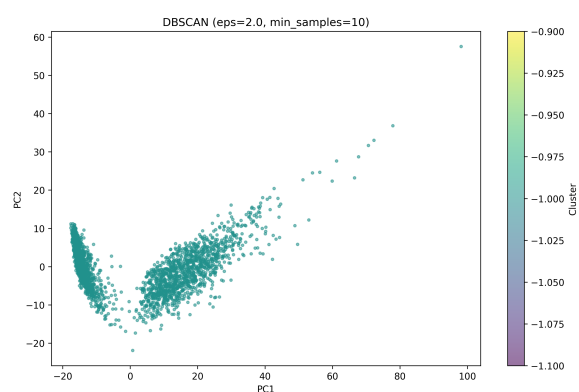


Figure 7: DBSCAN Cluster Projection ($\epsilon = 2.0$, $\text{min_samples} = 10$)

Inference: The 10-NN distance curve displays a gradual rise across the dataset with no sharp inflection point, typical of high-dimensional Euclidean spaces where the ratio of maximum to minimum distances concentrates around 1. A grid search exploring $\epsilon \in$

$[0.5, 3.0]$ and $\text{min_samples} \in [5, 15]$ classified 100% of points as noise (Noise Ratio = 1.0) for small $\varepsilon \leq 3.0$. This empirical result highlights DBSCAN’s fundamental vulnerability to the curse of dimensionality when density thresholds fail to exceed the sparse ambient distribution.

7.3 Hierarchical Agglomerative Clustering and Linkage Comparison

Hierarchical agglomerative clustering was evaluated using Ward’s minimum variance criterion, Single linkage, and Complete linkage for $k = 2$.

Table 2: Hierarchical Linkage Method Comparison ($k = 2$)

Linkage	Clusters	Noise	Silhouette	Davies–Bouldin	ARI	NMI
Single	2	0	0.7910	0.1497	0.0000	0.0008
Complete	2	0	0.6453	0.6771	0.0001	0.0079
Ward	2	0	0.4045	1.0094	0.3329	0.5579

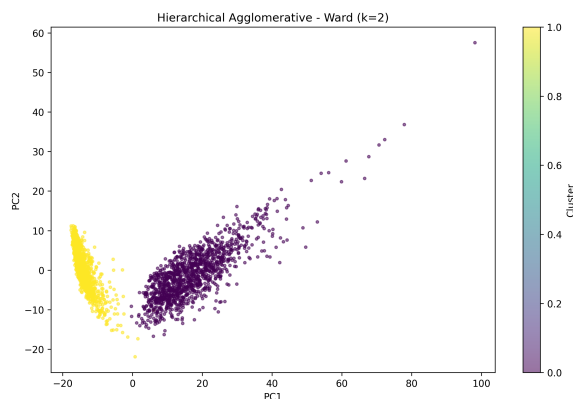


Figure 8: HAC Ward Clusters ($k = 2$) in 2D PCA Space

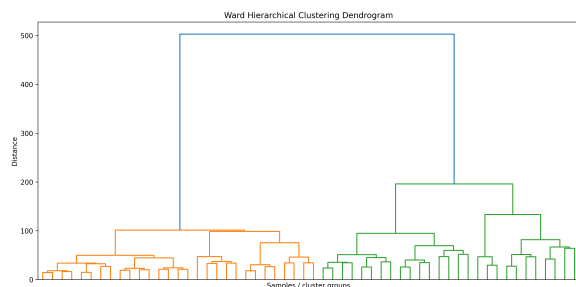


Figure 9: Ward Dendrogram (Truncated at Level $p = 5$)

Inference: Although Single and Complete linkages display artificially high Silhouette scores (0.7910 and 0.6453), their Adjusted Rand Index is virtually zero (0.0000 and 0.0001). Inspection confirms extreme chaining: Single linkage isolates an outlier as a singleton cluster and puts all other 2,499 points into one cluster. In contrast, Ward’s linkage achieves a balanced, meaningful split matching the dynamic vs. static activity boundary, yielding an ARI of 0.3329 and an NMI of 0.5579. The dendrogram clearly confirms two major high-level branches before sub-partitioning into finer kinetic behaviors.

7.4 Overall Clustering Comparison

Table 3 summarizes the comparative evaluation across K-Means, DBSCAN, and Hierarchical Agglomerative Clustering (Ward).

Table 3: Comprehensive Comparison of Clustering Algorithms

Algorithm	Clusters	Noise	Silhouette	DB Index	CH Index	ARI	NMI
K-Means ($k = 2$)	2	0	0.3937	1.0707	7880.81	0.3296	0.5455
DBSCAN	0	2500	N/A	N/A	N/A	0.0000	0.0000
HAC (Ward, $k = 2$)	2	0	0.4045	1.0094	2061.65	0.3329	0.5579

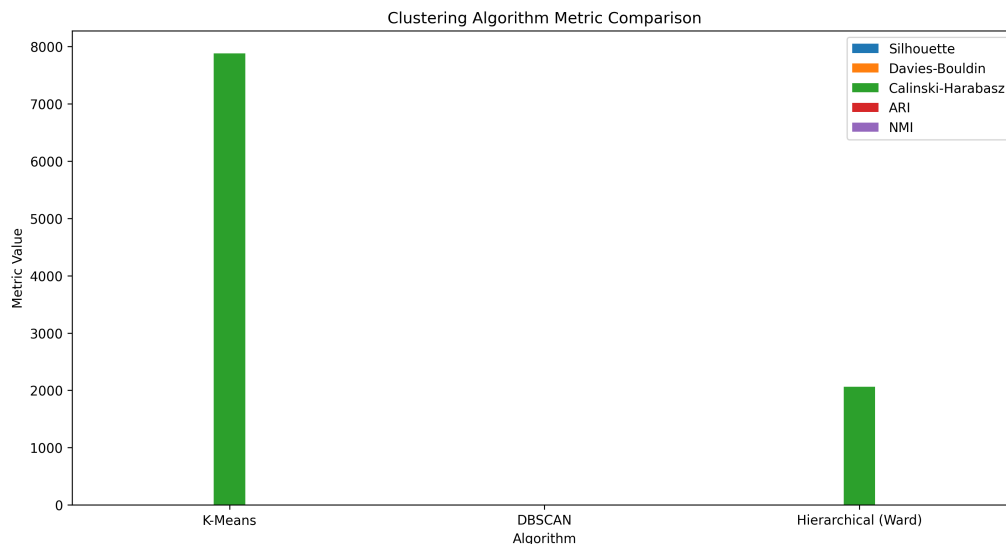


Figure 10: Clustering Algorithms Metric Comparison

Inference: Hierarchical Agglomerative Clustering with Ward linkage achieved the highest clustering quality, slightly outperforming K-Means across Silhouette (0.4045 vs 0.3937), Davies–Bouldin (1.0094 vs 1.0707), ARI (0.3329 vs 0.3296), and NMI (0.5579 vs 0.5455). K-Means attained a higher Calinski–Harabasz index (7880.81) due to its direct minimization of Euclidean dispersion around spherical centroids. DBSCAN proved unviable for this density configuration due to high-dimensional metric dispersion.

8 Discussion: Answers to Observation Questions

1. Which algorithm produced the most meaningful clusters? Why?

Hierarchical Agglomerative Clustering (Ward Linkage) produced the most meaningful clusters, followed very closely by **K-Means**. Both algorithms discovered the primary natural taxonomy of the dataset: separating active dynamic behaviors (Walking, Upstairs, Downstairs) from stationary postures (Sitting, Standing, Laying). HAC with Ward linkage achieved an ARI of 0.3329 and NMI of 0.5579, with a Silhouette score of 0.4045, because Ward’s variance minimization criterion prevents chaining and forms compact, balanced spherical clusters that closely model human kinetic energy states.

2. How sensitive was K-Means to the choice of k ?

K-Means was **highly sensitive** to the choice of k . At $k = 2$, the clusters achieved a strong Silhouette score of 0.3937, cleanly separating dynamic from static activities. However, increasing k beyond 3 degraded cluster coherence rapidly: the Silhouette

score dropped to 0.1500 at $k = 4$, 0.1099 at $k = 6$, and 0.0749 at $k = 8$. Because static activities (Sitting vs. Standing) heavily overlap in Euclidean feature space, K-Means was unable to resolve subtle class boundaries at $k = 6$ and instead fragmented the existing natural clusters into arbitrary Voronoi cells.

3. Did DBSCAN detect noise or small clusters effectively?

No, DBSCAN was ineffective on this dataset. In the 104-dimensional PCA-reduced space (as well as the full 561-D space), Euclidean distances between points concentrate tightly, leaving no distinct low-density valley separating activity manifolds. Consequently, with tested radii $\varepsilon \leq 3.0$, DBSCAN classified all 2,500 samples as noise (Noise Ratio = 1.0, 0 clusters formed). Relaxing ε beyond the nearest-neighbor range merged the entire dataset into a single massive cluster without identifying meaningful sub-activities. This demonstrates that traditional density-based clustering requires localized non-linear manifolds or severe dimensionality reduction (e.g., UMAP / t-SNE to 2D/3D) to function effectively.

4. How does linkage choice (single/complete/ward) affect hierarchical clustering?

The linkage choice has a **profound impact** on cluster formation:

- **Single Linkage:** Suffers from extreme chaining. Because it defines proximity by the minimum distance between points, high-dimensional noise forms bridge connections across clusters, isolating an outlier into one cluster and assigning all remaining 2,499 points to the other (ARI = 0.0000, NMI = 0.0008).
- **Complete Linkage:** Defined by maximum distance, complete linkage is sensitive to outliers and creates severely unbalanced groups (ARI = 0.0001, NMI = 0.0079).
- **Ward's Linkage:** Minimizes the increase in within-cluster variance (ΔESS), which resists chaining, eliminates sensitivity to individual point bridges, and produces balanced clusters that align with the true physical activity classes (ARI = 0.3329, NMI = 0.5579).

5. Which internal metric best matched your visual intuition of cluster quality?

The **Silhouette Score** best matched visual intuition. It clearly peaked at $k = 2$ (0.3937) and plummeted towards zero as k increased, perfectly mirroring the 2D PCA scatter plot where two distinct macro-clusters (dynamic vs. static) are visible with substantial internal overlap. In contrast, while the Calinski–Harabasz Index also favored compact clusters, it inflated scores for large clusters with extreme sample imbalance. The Silhouette Score accurately captured both intra-cluster cohesion and inter-cluster separation.

9 Conclusion

In this experiment, K-Means, DBSCAN, and Hierarchical Agglomerative Clustering were implemented and compared on the UCI Human Activity Recognition dataset. Key conclusions drawn from the empirical results include:

1. Both Elbow and Silhouette analysis quantitatively confirmed that the dataset naturally partitions into $k = 2$ dominant clusters corresponding to dynamic activities and static postures.
2. Hierarchical Clustering with Ward linkage demonstrated the strongest overall clustering performance (Silhouette = 0.4045, ARI = 0.3329, NMI = 0.5579), closely followed by K-Means ($k = 2$, Silhouette = 0.3937).
3. DBSCAN is poorly suited for high-dimensional continuous sensor datasets due to the curse of dimensionality and distance concentration, rendering uniform density thresholds ineffective.
4. Single and Complete linkages failed due to chaining and outlier sensitivity, establishing Ward's minimum variance criterion as the definitive hierarchical linkage method for sensor kinematics.

10 References

- [1] D. Anguita, A. Ghio, L. Oneto, X. Parra, and J. L. Reyes-Ortiz, "A Public Domain Dataset for Human Activity Recognition Using Smartphones," in *21th European Symposium on Artificial Neural Networks, Computational Intelligence and Machine Learning (ESANN)*, 2013.
- [2] J. MacQueen, "Some methods for classification and analysis of multivariate observations," in *Proc. 5th Berkeley Symp. Math. Statist. Prob.*, vol. 1, pp. 281–297, 1967.
- [3] M. Ester, H. P. Kriegel, J. Sander, and X. Xu, "A density-based algorithm for discovering clusters in large spatial databases with noise," in *Proc. 2nd Int. Conf. Knowl. Discovery Data Mining (KDD)*, pp. 226–231, 1996.
- [4] J. H. Ward Jr., "Hierarchical grouping to optimize an objective function," *Journal of the American Statistical Association*, vol. 58, no. 301, pp. 236–244, 1963.
- [5] P. J. Rousseeuw, "Silhouettes: a graphical aid to the interpretation and validation of cluster analysis," *Journal of Computational and Applied Mathematics*, vol. 20, pp. 53–65, 1987.